Assignment 10

1. What creative thinking technique involves individuals imagining themselves as part of the problem or situation they are trying to solve?

a. Fantasy analogy
b. Personal analogy
c. Symbolic analogy
d. CAD analogy

Answer: Personal analogy

2. Which of the following is NOT a typical prototyping method?

a. Additive methods
b. Subtractive methods
c. Forming methods
d. Lateral thinking

Answer: Lateral thinking

3. What is the correct sequence during a product development process?

a. Ideation → Concept Generation → Prototyping → Testing → Market Launch
b. Market Research → Concept Generation → Testing → Ideation → Prototyping
c. Concept Generation → Prototyping → Testing → Market Launch → Ideation
d. Ideation → Market Launch → Prototype → Testing → Concept Generation

Answer: Ideation → Concept Generation → Prototyping → Testing → Market Launch

4. Which type of reasoning involves forming general conclusions from specific observations?

a. Abductive reasoning
b. Deductive reasoning
c. Inductive reasoning
d. Analytical reasoning

Answer: Inductive reasoning

5. How does frugality in product innovation benefit organizations?

a. By increasing production costs to maximize resources
b. By minimizing waste and reducing costs through efficient use of resources
c. By prioritizing aesthetics over functionality
d. By focusing solely on rapid scaling without cost considerations

Answer: By minimizing waste and reducing costs through efficient use of resources

6. What is the primary purpose of a Proof of Concept (POC)?

a. To validate the feasibility and viability of a new idea or concept
b. To generate the final product for market launch
c. To secure funding for large-scale production
d. To build prototypes for user feedback

Answer: To validate the feasibility and viability of a new idea or concept

7. What does prototyping aim to achieve in the design process?

a. Finalizing the design for immediate market release
b. Creating models to test and improve design, functionality, and usability
c. Building large-scale production units for market distribution
d. Avoiding iterative development to save time

Answer: Creating models to test and improve design, functionality, and usability

8. What is the primary goal of Synectics in the design process?

a. Encouraging exploration and breaking conventional thinking
b. Generating a single, perfect solution to a problem
c. Reaching consensus among team members quickly
d. Applying detailed technical analysis to refine existing concepts

Answer: Encouraging exploration and breaking conventional thinking

9. How can entrepreneurs effectively collaborate with one another?

a. By franchising their mutual businesses
b. By sharing resources and distributing production responsibilities
c. By competing in the same market segments
d. All of the above

Answer: By sharing resources and distributing production responsibilities

10. Why is it important for designers to create prototypes before deploying a product?

a. To identify and resolve design flaws early in the process
b. To use prototypes as the final version for market release
c. To skip the need for market research and testing
d. To focus on aesthetics without user input

Answer: To identify and resolve design flaws early in the process